

Mixed-Precision Randomized Sketching for Scaled Dot-Product Attention

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Abstract. We propose a mathematically focused study of scaled dot-product attention under two simultaneous approximations: randomized feature sketching and mixed-precision arithmetic. The exact logit operator $d^{-1/2}QK^T$ is replaced by the sketched operator $d^{-1/2}QSS^TK^T$, where S is a random sketching matrix of smaller column dimension. The report isolates three levels of perturbation: the deterministic error in the logits, the induced perturbation of the rowwise softmax map, and the additional finite-precision error introduced when the sketched computation is carried out in low precision. The main analytical result is a clean error decomposition showing that the total output error is controlled by the sum of a sketching term and a rounding term. The accompanying experiments are designed as a validation of this decomposition rather than as a large-scale engineering benchmark.

Keywords: mixed-precision arithmetic, randomized numerical linear algebra, attention, transformers, sketching

1 Introduction

The purpose of this report is to study one analytically transparent component of the transformer architecture [1]: scaled dot-product attention. Two strands are particularly relevant. The first, rooted in randomized numerical linear algebra [2] [3], replaces the full attention matrix by a sketched surrogate [4] [5] [6]. The second, driven by hardware constraints, carries out the computation in reduced precision [7].

These two approximation strategies are typically studied in isolation. This report asks what happens when they are composed: when the query-key logit computation is approximated by a randomized sketch, can the subsequent computation be carried out in mixed precision with only a lower-order loss of accuracy?

The answer, as we show, is yes—and the mechanism is precise. Sketching reduces the effective dimension of the matrix product from d to s , which simultaneously controls the sketching error (via the intrinsic rank of the query-key subspace) and shrinks the finite-precision error (because the inner dimension of the product is smaller). The main result (Theorem 6) gives an explicit total error bound and highlights the condition $s \cdot u \ll \epsilon$ as the leading-order requirement for the finite-precision term to remain lower order when the accompanying norm factors are of comparable scale, where s is the

sketch dimension, u is the unit roundoff, and ϵ is the sketch tolerance. This condition provides a concrete design rule: sketch dimension and precision should be chosen jointly, not independently.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n \times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \rightarrow \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\text{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

Now let $S \in \mathbb{R}^{d \times s}$ with $s < d$. The sketched logit operator is

$$\tilde{\mathcal{L}}_S(Q, K) := d^{-1/2} Q S S^* K^*.$$

The operator $P_S := S S^*$ is positive semidefinite with rank at most s , but it is not in general an orthogonal projector unless the columns of S are orthonormal. The corresponding sketched attention output is

$$\tilde{\mathcal{A}}_S(Q, K, V) := \Sigma(\tilde{\mathcal{L}}_S(Q, K))V.$$

Thus the basic approximation step is the replacement of inner products $\langle q_i, k_j \rangle$ by sketched inner products $\langle S^* q_i, S^* k_j \rangle$.

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map.

Proposition 1 (Rowwise softmax is Hilbert–Schmidt Lipschitz). Let $L, M \in \mathbb{R}^{n \times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \leq \|L - M\|_{\text{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n \times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \leq \|L - M\|_{\text{HS}} \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the softmax map on vectors with Jacobian $D\sigma(x) = \text{Diag}(p) - pp^T$ where $p = \sigma(x)$. For any $z \in \mathbb{R}^n$, $z^T D\sigma(x) z = \sum_i p_i z_i^2 - (\sum_i p_i z_i)^2 \leq \|z\|_2^2$, so $\|D\sigma(x)\|_2 \leq 1$ for all x . The mean value theorem gives $\|\sigma(x) - \sigma(y)\|_2 \leq \|x - y\|_2$; applying this row by row and squaring yields the HS bound. The second assertion follows from $\|XV\|_{\text{HS}} \leq \|X\|_{\text{HS}} \|V\|_2$. $\square$

The next proposition isolates the deterministic sketching error. Since attention logits are bilinear forms $q_i^* k_j$, the relevant notion is that S approximately preserves inner products on the row space of Q and K .

Proposition 2 (Deterministic sketching bound). Let $U \subseteq \mathcal{H}_d$ be the joint row space of Q and K , equivalently the subspace spanned by the columns of $[Q^T \ K^T]$, and let $r := \dim U$. Assume that S is an ϵ -subspace embedding for U , meaning

$$|\langle S^* u, S^* v \rangle - \langle u, v \rangle| \leq \epsilon \|u\|_2 \|v\|_2 \quad \text{for all } u, v \in U.$$

Equivalently, if $U_0 \in \mathbb{R}^{d \times r}$ is any orthonormal basis matrix for U , then $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$. Then

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning the joint row space of Q and K . Write $Q = Q_U U_0^T$ and $K = K_U U_0^T$, where $Q_U = Q U_0 \in \mathbb{R}^{n \times r}$ and $K_U = K U_0 \in \mathbb{R}^{n \times r}$. Then

$$QK^* - QSS^*K^* = Q_U(I_r - U_0^T S S^* U_0)K_U^T.$$

The subspace-embedding condition is exactly $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$, so

$$\|QK^* - QSS^*K^*\|_{\text{HS}} \leq \|Q_U\|_{\text{HS}} \|I_r - U_0^T S S^* U_0\|_2 \|K_U^T\|_2 \leq \epsilon \|Q\|_{\text{HS}} \|K\|_2,$$

where we used $\|Q_U\|_{\text{HS}} = \|Q\|_{\text{HS}}$ and $\|K_U\|_2 = \|K\|_2$. Multiplying by $d^{-1/2}$ gives the logit bound. The attention-output bound follows from Proposition 1. $\square$

The proposition should be read as a conditional stability statement: it separates the attention-stability argument from the probabilistic question of when the sketch approximately preserves inner products on the relevant subspace.

The next result gives this probabilistic guarantee for Gaussian sketches. A scaled Gaussian matrix $S \in \mathbb{R}^{d \times s}$ with entries $\mathcal{N}(0, 1/s)$ is a subspace embedding for any fixed r -dimensional subspace with high probability, requiring $s = O(r/\epsilon^2)$ —not $O(d/\epsilon^2)$.

Proposition 3 (Probabilistic subspace embedding guarantee). Let $U \subseteq \mathcal{H}_d$ be an r -dimensional subspace. Let $S \in \mathbb{R}^{d \times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For

$\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, if

$$s \geq C \frac{r + \ln(1/\delta)}{\epsilon^2},$$

for a universal constant C , then with probability at least $1 - \delta$, S is an ϵ -subspace embedding for U . If U contains the joint row space of Q and K , then with the same probability,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d \times r}$ be a matrix whose columns form an orthonormal basis for U . Then S is an ϵ -subspace embedding for U if and only if $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$. The matrix $U_0^T S S^* U_0 = (S^* U_0)^* (S^* U_0)$ has the same distribution as $G^* G/s$ where $G \in \mathbb{R}^{s \times r}$ has i.i.d. standard normal entries. Standard JL concentration [2] gives $\Pr(\|G^* G/s - I_r\|_2 > \epsilon) \leq 2 \exp(-c s \epsilon^2 + r)$ for a universal constant $c > 0$, provided s is large enough relative to r/ϵ^2 . The stated condition with an appropriate C ensures the failure probability is at most δ . The attention error bound follows from Proposition 2. $\square$

The key feature is the dependence on r rather than d : when the query-key rows lie near a low-dimensional subspace, a much smaller sketch suffices.

We now turn to the finite-precision error. The standard Higham-type forward-error analysis [8] for the matrix product AB with $A \in \mathbb{R}^{n \times s}$, $B \in \mathbb{R}^{s \times n}$ gives

$$|\widehat{AB} - AB| \leq \gamma_s |A| |B|, \quad \gamma_s := \frac{su}{1 - su},$$

where u is the unit roundoff. For the sketched logit product $d^{-1/2}(QS)(KS)^*$, the inner dimension is s , so the finite-precision error is proportional to $s \cdot u$ —not $d \cdot u$ as it would be for the full product QK^* .

Proposition 4 (Finite-precision error for sketched logits). Let $\hat{\mathcal{L}}_S(Q, K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u , and let $\gamma_s := su/(1 - su)$ for $su < 1$. Then

$$\|\tilde{\mathcal{L}}_S(Q, K) - \hat{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}}}{\sqrt{d}}.$$

Consequently,

$$\|\tilde{\mathcal{A}}_S(Q, K, V) - \hat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}} \|V\|_2}{\sqrt{d}}.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n \times s}$ and $B = KS \in \mathbb{R}^{n \times s}$. The standard forward-error bound for matrix multiplication [8] gives

$$|\widehat{AB}^* - AB^*| \leq \gamma_s |A| |B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm and applying $\lrcorner$

↳ the Cauchy–Schwarz inequality for matrices,

$$\|\widehat{AB}^* - AB^*\|_{\text{HS}} \leq \gamma_s \| |A| |B^*| \|_{\text{HS}} \leq \gamma_s \|A\|_{\text{HS}} \|B\|_{\text{HS}}.$$

Multiplying by $d^{-1/2}$ yields the logit bound. The attention-output bound follows from Proposition 1. $\square$

Remark 5 (Scope of the finite-precision bound). The bound in Proposition 4 models the error in the final sketched logit product $(QS)(KS)^*$, assuming that QS and KS are given exactly. The experiments measure a broader implementation error: Q , K , V , and S are first quantized to the storage dtype, and the sketched factors QS and KS are then formed from those quantized tensors before the final logit product. Thus the measured E_{fp} includes storage quantization and low-precision projection effects in addition to the multiplication-only term analyzed in the proposition. $\lrcorner$

The factor γ_s scales linearly with s : compared to an unsketched product with inner dimension d , sketching changes the leading roundoff coefficient from $O(d \cdot u)$ to $O(s \cdot u)$. This is the formal statement of the intuition that “compressing first makes low precision safer.” Combining the probabilistic subspace embedding guarantee with the finite-precision error analysis yields the main result.

Theorem 6 (Mixed-precision sketched attention). Under the hypotheses of Propositions 3 and 4, with probability at least $1 - \delta$,

$$\|\mathcal{A}(Q, K, V) - \widehat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \left(\epsilon \|Q\|_{\text{HS}} \|K\|_2 + \gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}} \right) \|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error.

Proof. By Proposition 1 and the triangle inequality,

$$\|\mathcal{A} - \widehat{\mathcal{A}}_S\|_{\text{HS}} \leq (\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}} + \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}) \|V\|_2.$$

Apply Proposition 2 to the first term and Proposition 4 to the second. $\square$

Remark 7 (Lower-order regime). Theorem 6 isolates the roundoff coefficient $\gamma_s = O(s \cdot u)$ explicitly. If, in addition, the norm factors satisfy a scale comparison such as

$$\|QS\|_{\text{HS}} \|KS\|_{\text{HS}} = O(\|Q\|_{\text{HS}} \|K\|_2),$$

then $s \cdot u \ll \epsilon$ is the leading-order requirement for the finite-precision term to remain lower order than the sketching term. $\lrcorner$

Remark 8 (Design rule). Theorem 6 yields a concrete joint design rule. For float16 ($u_{16} \approx 5 \times 10^{-4}$) and typical sketch dimensions $s \leq 64$, we have $s \cdot u_{16} \leq 0.032$, which is smaller than typical sketch tolerances $\epsilon \geq 0.1$. For float32 ($u_{32} \approx 6 \times 10^{-8}$), the condition is essentially always satisfied. For float8 E4M3 ($u_{\text{fp8}} \approx 0.0625$), even $s = 16$ gives $s \cdot u_{\text{fp8}} \approx 1.0$, which is comparable to ϵ —so the finite-precision term is no longer clearly lower order. The rule also reveals a regime to avoid: if s is chosen too large relative to $1/u$, the finite-precision error can overtake the sketching error, defeating the purpose of mixed precision. $\lrcorner$

4 Numerical Experiments

The experiments probe the decomposition of Theorem 6 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

- (1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);
- (2) $\tilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;
- (3) $\hat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching error is $E_{\text{sk}} := \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$ and the finite-precision error is $E_{\text{fp}} := \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$. The total error is $E_{\text{tot}} := \|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$. The same decomposition is recorded for the attention output. We test four precision configurations: `fp32` (all stages in float32), `full_fp16` (storage, projection, final logit product, and softmax in float16), `mixed_fp16` (storage and projection in float16, followed by a float32 final logit product and float32 softmax), and `fp8_store` (storage and projection in float8 E4M3, followed by a float32 final logit product and float32 softmax).

The sketch $S \in \mathbb{R}^{d \times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds. Data and sketch seeds are independent. The full numerical results are available in the committed CSV file `results/attention_sweep.csv`.

Since the synthetic Q and K are i.i.d. Gaussian, their joint row space is typically full-dimensional, so for $d = 64$ and $s \leq 32$ the subspace-embedding regime from Proposition 3 is not expected to hold. The experiment should therefore be read as an intentionally under-sketched stress test of the error decomposition, not as evidence that such small feature sketches preserve attention accurately.

Table 9 reports absolute HS errors of the logits for $n = 256$, $d = 64$, together with the ratio $E_{\text{sk}}/E_{\text{fp}}$; the output errors showed the same qualitative ordering and are omitted.

Three observations stand out. First, for `fp32` and `fp16`, $E_{\text{sk}}/E_{\text{fp}}$ is consistently large (10^3 – 10^6), confirming that the dominant error source is the randomized sketch. The relative sketching error exceeds 140%, so this regime is not a good attention approximation; rather, it is a deliberate stress test in which the sketching floor is already very high, while the relative finite-precision error stays below 0.16%. Second, `full_fp16` and `mixed_fp16` have finite-precision errors of the same order, consistent with Proposition 4: storage quantization (the $s \cdot u$ factor) dominates any benefit from higher-precision accumulation. Third, `fp8` storage substantially increases the finite-precision error relative to `fp16` (from ~ 0.3 to ~ 30 in absolute terms, or from $\sim 0.1\%$ to $\sim 10\%$ relative), but the sketching error still dominates by a factor of ~ 14 – 15 . That is, `fp8` storage degrades accuracy noticeably compared to `fp16`, yet in the tested synthetic regime the total error remains sketching-dominated. A regime where finite precision dominates would require even lower precision or larger s ; the current experiments do not reach it. Thus the results support the error decomposition and illustrate the degradation predicted by Theorem 6 for increasing $s \cdot u$, but they do not demonstrate a high-quality efficient-attention approximation or a full inversion of the error ratio.

Config	s	E_{sk}	E_{fp}	E_{tot}	$E_{\text{sk}}/E_{\text{fp}}$
fp32	8	7.22e2	1.2e-4	7.22e2	5.9e6
	16	5.03e2	1.1e-4	5.03e2	4.4e6
	24	4.27e2	9.4e-5	4.27e2	4.5e6
	32	3.70e2	9.1e-5	3.70e2	4.1e6
full_fp16	8	7.22e2	4.1e-1	7.22e2	1.8e3
	16	5.03e2	2.9e-1	5.03e2	1.8e3
	24	4.27e2	2.6e-1	4.27e2	1.7e3
	32	3.70e2	2.3e-1	3.70e2	1.6e3
mixed_fp16	8	7.22e2	3.8e-1	7.22e2	1.9e3
	16	5.03e2	2.6e-1	5.03e2	1.9e3
	24	4.27e2	2.4e-1	4.27e2	1.8e3
	32	3.70e2	2.1e-1	3.70e2	1.8e3
fp8_store	8	7.22e2	5.0e1	7.21e2	1.4e1
	16	5.03e2	3.4e1	5.01e2	1.5e1
	24	4.27e2	3.0e1	4.26e2	1.4e1
	32	3.70e2	2.6e1	3.70e2	1.4e1

Table 9 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds).
 $E_{\text{sk}} = \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{fp}} = \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{tot}} = \|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$; $\|\mathcal{L}\|_{\text{HS}} \approx 256$.

5 Outlook

The present report studies only the feature-sketch model $QK^* \approx QSS^*K^*$. This is analytically transparent, but it is not the strongest empirical setting for efficient attention: it compresses the model dimension d , whereas the main computational bottleneck in long-sequence transformers is usually the sequence dimension n , and on generic Gaussian data the relevant row space is essentially full-dimensional. The most promising next step is therefore to move from feature sketching to random-feature approximation of the softmax kernel, following Performer-style constructions [6]. In that setting one approximates

$$\exp(q^*k/\sqrt{d}) \approx \phi(q)^*\phi(k),$$

which leads to a kernelized attention surrogate of the form

$$\tilde{A}(Q, K, V) = \tilde{D}^{-1}\Phi(Q)\Phi(K)^*V.$$

This would support a more compelling error decomposition,

$$\|A - \hat{A}\| \leq \|A - \tilde{A}\| + \|\tilde{A} - \hat{A}\|,$$

where the first term is the random-feature approximation error and the second is the additional finite-precision error. A natural design rule would then compare the random-feature dimension m with the unit roundoff u through a condition of the form $mu \ll \epsilon_{\text{rf}}$, where ϵ_{rf} is the random-feature error floor.

5.1 Random-Feature Prototype

The repository now contains a prototype implementation of this pivot. It uses a positive orthogonal random-feature map modeled on Performer-style constructions:

the projection matrix is assembled from block-orthogonal Gaussian factors, and the exponential feature map is stabilized by subtracting a querywise maximum for queries and a global maximum for keys before exponentiation. This keeps the map positive while reducing overflow risk in finite precision.

The prototype evaluates the decomposition

$$\|A - \hat{A}\| \leq \|A - \tilde{A}\| + \|\tilde{A} - \hat{A}\|,$$

at three levels: the unnormalized softmax kernel, the normalized attention weights, and the final attention output. Here A denotes exact softmax attention, $\tilde{A}$ is the same random-feature approximation evaluated in `float64`, and $\hat{A}$ is the corresponding low-precision implementation. The first term is therefore the intrinsic random-feature approximation floor, while the second isolates the additional precision loss for a fixed feature map.

To make the experimental regimes more informative than the current full-rank Gaussian feature-sketch test, the prototype supports three input families: (i) i.i.d. Gaussian Q, K, V as a baseline, (ii) low-rank synthetic activations of the form $Q = ZA + \sigma N_Q$, $K = ZB + \sigma N_K$, and (iii) query, key, and value tensors extracted from a selected head of a pretrained transformer layer on real text. This creates a direct path from mathematically controlled synthetic models to structured activations arising in an actual language model.

That pivot allows experiments on more realistic data while keeping the mixed-precision question central. Other possible extensions are Nyström-type approximations [5], structured random matrices in place of Gaussian sketches, and an operator-theoretic treatment for Hilbert–Schmidt limits under compactness or spectral-decay assumptions.

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